

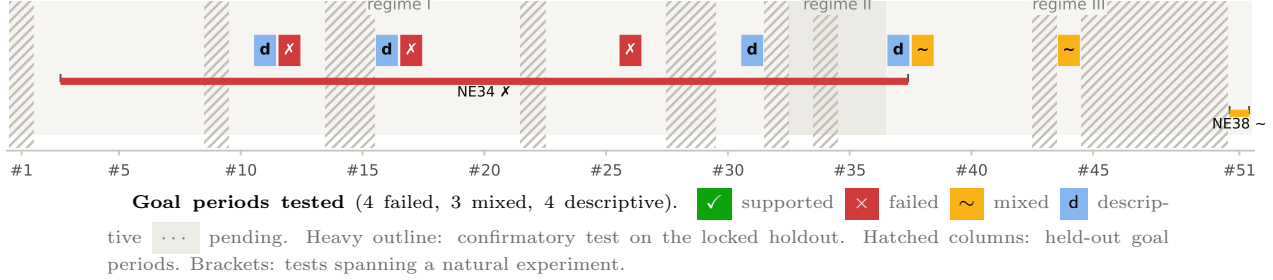
H10: Goals are Legendre pushes: a swarm's response to an assigned goal is predictable from its unforced fluctuations

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 1/5 Usefulness 1.5/5

Bottom line. Goals are quenches toward a common target, not Legendre pushes: free-week fluctuations do not predict who follows, in both embedding models.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

The question is whether a swarm's unforced fluctuations predict how it lines up behind an assigned goal. The state is each agent's 30-min statement embedding, projected on the goal direction $\hat{g}$. Three free $\rightarrow$ assigned pairs exist (#11 $\rightarrow$ #12, #16 $\rightarrow$ #17, #37 $\rightarrow$ #38). In each pair one field λ fits the mean shift, and the model predicts the rest. Alignment rises 1.4–3.1 free-week SDs, outside linear response. **The core test fails:** fluctuations do not predict who moves ($r = -0.37, -0.69, +0.20$; Stouffer $p = 0.89$). **Round 1b** uses shared goal vectors, a second embedding model, dedupe and style residualization. The test fails in all 8 input configurations. Variance along $\hat{g}$ grows, and transverse variance holds. In a new native test a reassigned goal moves one agent 10–20 SDs, beyond any point its unforced content visits. An agent-set goal moves agents, but again not in fluctuation order. The holdout is frozen and has not run.

PHYSICAL MODEL

Model 11: mean-field $O(n)$ in $n = 32$ whitened embedding dimensions. $\mathbf{s}_i$ are agent i 's unit statement vectors. $y = \hat{\mathbf{z}} \cdot \hat{\mathbf{g}}$ is the alignment with the goal field direction. x_i is its 30-min window mean, with free-week mean μ_i , noise-corrected variance $\kappa_{2,i}$ and third cumulant $\kappa_{3,i}$. The goal is a linear field $\lambda = \beta h$ along $\hat{g}$. K is the cumulant generating function, J_0 the mean-field coupling, g the loop gain and C_F the free-week window covariance.

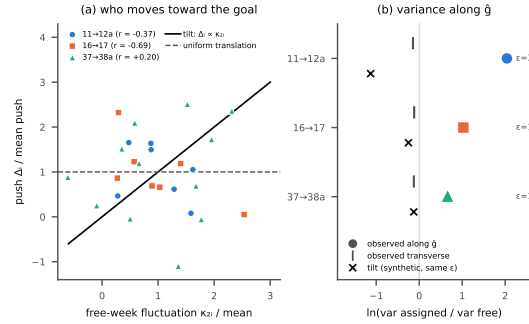
$$P_A(\mathbf{s}) \propto P_F(\mathbf{s}) e^{\lambda \sum_i \hat{\mathbf{g}} \cdot \mathbf{s}_i}, \quad G = F - \lambda m$$

$$K_A(t) = K_F(t + \lambda) - K_F(\lambda) \Rightarrow \kappa_n^A = K_F^{(n)}(\lambda)$$

$$\Delta \mu_i = \lambda \kappa_{2,i} + \frac{\lambda^2}{2} \kappa_{3,i}, \quad \kappa_{2,i}^A = \kappa_{2,i} + \lambda \kappa_{3,i}$$

$$\chi_{\text{swarm}} = \chi_0 / (1 - \beta J_0 \chi_0), \quad g = \beta J_0 \chi_0 = 1 - V_{\text{indep}} / V$$

$$\Delta \mu = \lambda C_F \hat{g}$$



Observables. (a) Each agent's push toward the assigned goal against its free-week fluctuation along $\hat{g}$, both divided by the pair mean. The tilt predicts the solid line, uniform translation the dashed one. The regime-I pairs slope the wrong way. (b) The change in variance along $\hat{g}$ (colored), along transverse directions (bars), and for an exact tilt of a bounded mean-field swarm at the same push ($\times$, synthetic). The goal disperses the swarm along $\hat{g}$; it does not narrow it.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

The promise was to read a swarm's steerability from its unsteered fluctuations. That promise does not hold here. A goal change is a large kick (median about 2.5 SDs), not a small field. The agents that wander most are not the agents that follow most. An assigned goal acts as a common target and a new collective mode, not as a gentle tilt of existing preferences. Do not forecast compliance from pre-goal behavior: the fluctuation–response correlation runs from -0.69 to $+0.20$. The instrument (goal-direction projection, noise-corrected variance) is cheap on any agent log. The fluctuation–response physics it tests is a curiosity in this village.

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RESULTS VS PREDICTIONS

Prediction	Round 1 $\rightarrow$ 1b (bge / gte)	Verdict
P1 fluctuations order who moves	r $-0.37, -0.69, +0.20 \rightarrow$ same / $\times$ $-0.53, -0.49, -0.09$; 8/8 configs fail	$\times$
Push $\Delta\bar{\mu} > 0$	3/3 pairs; 19/19 kickoffs $\rightarrow$ same $\checkmark$ in all configs	$\checkmark$
P2 variance by the tilt	untestable ($\varepsilon > 0.34$); gte n/a #37 $\rightarrow$ #38 would pass (ε 0.69)	n/a
Cross-kickoff man > 0	Spear- $-0.56 \rightarrow -0.56 / -0.59$	$\times$
G44 assigned vs own-choice room	#best ε 0.97 / 1.55; #rest none; $\sim$ gap +0.17	$\sim$
NE38 one agent re-signed	ε 10 / 20; post mean beyond the pre support	(tilt $\times$)
G26 agent-set goal	push $\varepsilon \approx 1$; P1 r $-0.14 / -0.63$	$\times$

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	0	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

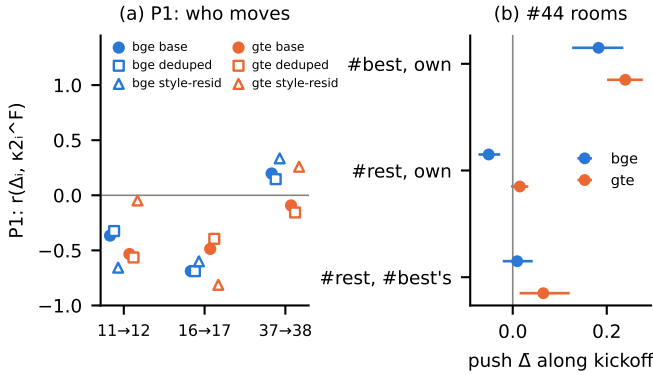
CAVEATS

$\hat{g}$ is a text-embedding proxy for the field. gte shrinks the regime-III push and makes its variance tilt-compatible, so the regime-III result depends on the model. Regime I has $N = 7$. NE38's concurrent control is one incumbent, which drifts 1–3 SDs along its own goal. The #44 assigned room has 4 agents and three days. The natives are exploratory. #22 $\rightarrow$ #23 and #31 $\rightarrow$ #32 remain blind.

ROUND 2

Re-map goals as a two-state (on/off-goal) occupancy. Test field against contagion onsets (H10-R1/R2). Run `confirm.py` only after Vivian's sign-off.

- **Essence.** How does a new goal propagate: as a field acting on everyone at once, or as contagion from early adopters?
- **H10-R1** Adoption onsets after a kickoff: a field predicts simultaneous onsets; contagion predicts onsets ordered by exposure to early adopters.
- **H10-R2** Goals act on a two-state on-goal/off-goal occupancy (HH130).
- **H10-R3** During the week, goal-specific content relaxes back toward family-prior directions (E4).



Second figure. (a) P1 per pair under every round-1b input (blue bge, orange gte; filled base, squares deduped, triangles style-residualized). The regime-I pairs stay negative. (b) #44 native: the assigned room moves along its own kickoff, and the self-chosen room does not (90% CIs).

GOAL PERIODS IN DETAIL

G11 d 1b: descriptive (both models)
 G12 $\times$ 1b: failed (both models)
 G16 d 1b: descriptive (both models)
 G17 $\times$ 1b: failed (both models)
 G26 $\times$ failed (round 1b native; both models)
 G31 d 1b: descriptive (both models)
 G37 d 1b: descriptive (both models)
 G38 $\sim$ 1b: mixed (bge mixed, gte failed)
 G44 $\sim$ mixed (round 1b native; model-dependent)
 NE34 $\times$ 1b: failed (both models)
+1 more periods: see the card.

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